Mutual Fund Analytics Platform

Day 2 – Data Cleaning + SQL Database Design

Internship Report submitted to

Bluestock Fintech

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This report covers data cleaning, validation, star schema design, SQLite warehouse creation, and analytical query development for the Mutual Fund Analytics Platform.

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Abstract

This report documents Day 2 activities of the Mutual Fund Analytics Platform internship project at Bluestock Fintech. The work involved cleaning and validating all ten mutual fund datasets, designing a SQLite star schema data warehouse, loading cleaned data into dimensional and fact tables, and developing ten analytical SQL queries for business insights.

All datasets passed validation checks with zero AMFI mismatches, zero duplicates, and zero foreign key violations. The warehouse was loaded successfully and is ready for dashboard and reporting use.

Project Overview

Organization: Bluestock Fintech **Date:** 24/06/2026

Project: Mutual Fund Analytics Platform **Day:** Day 2

Objectives

- Clean and validate all ten mutual fund datasets
- Standardize date formats and apply business rules
- Design and build a SQLite star schema data warehouse
- Load all cleaned datasets into dimensional and fact tables
- Develop analytical SQL queries for business insights

Work Completed

- Cleaned all 10 datasets – removed duplicates, fixed formats, handled missing values
- Standardized date columns across all datasets to `YYYY-MM-DD` format
- Validated AMFI codes and transaction categories
- Checked expense ratios and created anomaly flags
- Designed and implemented SQLite star schema
- Created dimension tables: `dim_fund`, `dim_date`
- Created fact tables: `fact_nav`, `fact_transactions`, `fact_performance`, `fact_aum`
- Loaded all cleaned data into the warehouse successfully
- Verified referential integrity across all fact tables
- Developed 10 analytical SQL queries
- Generated data dictionary and validation report

Project Folder Structure

Folder / File	Purpose
`data/raw/`	Original source datasets
`data/processed/`	Cleaned and validated datasets
`scripts/clean_all_data.py`	Cleans and validates all 10 datasets
`scripts/load_sqlite.py`	Loads cleaned data into SQLite warehouse
`sql/schemas.sql`	Star schema DDL definitions
`sql/queries.sql`	Analytical SQL queries
`reports/data_dictionary.md`	Column-level data dictionary
`reports/validation_report.md`	Validation summary and findings
`bluestock_mf.db`	SQLite database file

Dataset Overview

All ten datasets were cleaned and saved to `data/processed/` as new CSV files.

No.	Cleaned Dataset	Shape (Rows × Cols)
1	`01_fund_master_clean.csv`	40 × 15
2	`02_nav_history_clean.csv`	46,000 × 3
3	`03_aum_by_fund_house_clean.csv`	90 × 5
4	`04_monthly_sip_inflows_clean.csv`	48 × 7
5	`05_category_inflows_clean.csv`	144 × 3
6	`06_industry_folio_count_clean.csv`	21 × 7
7	`07_scheme_performance_clean.csv`	40 × 21
8	`08_investor_transactions_clean.csv`	32,778 × 13
9	`09_portfolio_holdings_clean.csv`	322 × 8
10	`10_benchmark_indices_clean.csv`	8,050 × 3

Data Cleaning and Validation

Cleaning Steps Applied

- Converted all date columns to `YYYY-MM-DD` datetime format
- Removed duplicate records across all datasets
- Validated AMFI codes for consistency between fund master and NAV history
- Validated transaction category values against allowed categories
- Checked expense ratios for out-of-range values
- Identified 12 expected null values in SIP YoY growth for the first year, where no prior-year comparison exists.
- Added anomaly flags for SIP YoY growth, return values, and expense ratio checks.
- Verified column data types and cast where necessary

Validation Checks Performed

- AMFI code cross-reference between fund master and all linked datasets
- Duplicate row detection using full row comparison

- Null value check on all primary key and foreign key columns
- Range validation on numeric fields (expense ratio, exit load, NAV)
- Referential integrity check after warehouse loading

SQLite Star Schema Design

Schema Overview

A star schema was designed to support efficient analytical querying. The schema separates descriptive attributes into dimension tables and measurable facts into fact tables.

Table	Type	Description
`dim_fund`	Dimension	Fund metadata: name, house, category, risk, benchmark
`dim_date`	Dimension	Date attributes: year, month, quarter, day of week
`fact_nav`	Fact	Daily NAV values linked to fund and date dimensions
`fact_transactions`	Fact	Investor transaction records with amount and type
`fact_performance`	Fact	Scheme-level return and risk metrics
`fact_aum`	Fact	AUM figures by fund house and date

Design Decisions

- `dim_fund` uses `amfi_code` as the natural key
- `dim_date` was generated to cover the full date range of all datasets
- Fact tables reference the relevant dimensions based on their grain(level of detail).
- Schema is normalized at the dimension level to avoid redundancy

Database Loading and Verification

Loading Process

- All cleaned CSV files were loaded into the SQLite database `bluestock_mf.db`
- Dimension tables were loaded first to satisfy foreign key constraints
- Fact tables were loaded after dimensions were fully populated
- Row counts were verified after each table load

Database Row Counts

Table	Type	Row Count
`dim_fund`	Dimension	40
`dim_date`	Dimension	1,610
`fact_nav`	Fact	46,000
`fact_transactions`	Fact	32,778
`fact_performance`	Fact	40
`fact_aum`	Fact	90

Verification Results

- All row counts matched the cleaned CSV file shapes exactly
- No records were lost during the loading process
- Foreign key constraints were satisfied across all fact tables
- Database file `bluestock_mf.db` was created successfully

Analytical Queries

Ten analytical SQL queries were developed and stored in `sql/queries.sql`. These queries are designed to support business insights and dashboard development.

No.	Query Description
1	Top 5 individual mutual fund schemes ranked by scheme-level AUM.
2	Average NAV per month
3	SIP YoY growth
4	Transactions by state
5	Funds with expense ratio below 1%
6	Top 5 funds by 5-year return
7	AUM by fund house
8	Monthly redemption trends
9	Risk grade distribution
10	Top investing states

Data Quality Summary

Validation Check	Result
AMFI code mismatches	0
Duplicate records after cleaning	0
Foreign key violations	0
Null fund_id in fact tables	0
Row loss during warehouse loading	0

Dataset	Issues Found	Status After Cleaning
Fund Master	None	Clean
NAV History	None	Clean
AUM by Fund House	None	Clean
Monthly SIP Inflows	12 null values in `yoy_growth_pct`	The missing values are expected because the first comparison periods do not have data from the previous year for YoY growth calculation.
Category Inflows	None	Clean
Industry Folio Count	None	Clean
Scheme Performance	None	Clean
Investor Transactions	None	Clean
Portfolio Holdings	None	Clean
Benchmark Indices	None	Clean

Key Findings

- All 10 datasets passed validation checks with zero critical issues
- NAV history contains 46,000 records covering a full multi-year period
- Investor transactions dataset contains 32,778 records across multiple cities and investor profiles
- Warehouse loading preserved all records with no data loss
- Star schema design supports efficient joins and analytical queries
- Scheme-level AUM analysis identifies the largest individual funds, while fund-house AUM analysis compares total AMC-level assets.
- SIP inflow data supports month-on-month and year-on-year growth analysis
- Expense ratio filtering allows direct comparison of scheme costs
- Referential integrity is fully maintained across all fact tables
- The database `bluestock_mf.db` is ready for dashboard and reporting use

Conclusion

Day 2 of the Mutual Fund Analytics Platform project has been completed successfully. All ten datasets were cleaned, validated, and loaded into a structured SQLite star schema warehouse. Data quality checks confirmed zero mismatches, zero duplicates, and zero foreign key violations. Ten analytical queries were developed to support business intelligence use cases.